

## Problem Set #1-- Answer Key

Q1)

a)

$$i) \text{ Prob}(Y=1|X=0) = \frac{\text{Prob}(Y=1, X=0)}{\text{Prob}(X=0)} = \frac{0.4}{0.5} = 0.8$$

$$E[Y|X=0] = \text{Prob}(Y=1|X=0) = 0.8$$

$$\text{Prob}(Y=1|X=1) = \frac{\text{Prob}(Y=1, X=1)}{\text{Prob}(X=1)} = \frac{0.3}{0.5} = 0.6$$

$$E[Y|X=1] = \text{Prob}(Y=1|X=1) = 0.6$$

ii)

$$\text{Var}(Y|X=0) = \text{Prob}(Y=1|X=0) \times \text{Prob}(Y=0|X=0) = 0.16$$

$$\text{Var}(Y|X=1) = \text{Prob}(Y=1|X=1) \times \text{Prob}(Y=0|X=1) = 0.24$$

b)

Let

$$m(x) = 0.8 - 0.2x, \quad \text{then}$$

$$u = y - (0.8 - 0.2x)$$

$$\begin{aligned} E[u|x=0] &= (1-0.8)(0.8) + (0-0.8)(0.2) \\ &= (0.2)(0.8) + (-0.8)(0.2) = 0 \end{aligned}$$

and

$$\begin{aligned} E[u|x=1] &= (1-0.6)(0.6) + (0-0.6)(0.4) \\ &= (0.4)(0.6) + (-0.6)(0.4) = 0 \end{aligned}$$

By LIE

$$\begin{aligned} E[u] &= E[E[u|x]] \\ &= 0(0.5) + 0(0.5) = 0 \end{aligned}$$

Also,

$$\begin{aligned} E[m(x)u] &= E[E[m(x)u|x]] \\ &= E[m(x)E[u|x]] \\ &= 0 \end{aligned}$$

So

$$\text{Var}(y) = \text{Var}(m(x)) + \text{Var}(u)$$

$$\text{Var}(u) = \text{Var}(y) - \text{Var}(W(X))$$

$$\text{Var}(u) = \text{Var}(y) - 0.04 \text{Var}(X)$$

$$\text{Var}(u) = (0.3)(0.7) - (0.04)(0.5)(0.5)$$

$$\text{Var}(u) = 0.2$$

(c)

$$\hat{\beta}_0 = \frac{\sum_{i=1}^{n_1} y_i}{n_1}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n_2} y_i}{n_2} - \frac{\sum_{i=1}^{n_1} y_i}{n_1}$$

where  $n_1 + n_2 = n$

$n_1 \rightarrow$  obs with  $X=0$

$n_2$  obs with  $X=1$

$$\text{Var}(y|X) = \text{Var}((m(X) + u)|X)$$

$$\text{Var}(y|X) = \text{Var}(u|X)$$

$$\text{Var}(u|X=0) \neq \text{Var}(u|X=1)$$

So, there is heteroskedasticity  
d)

$$E[X|Y=0] = \text{Prob}(X=1|Y=0) = \frac{0.2}{0.3} = \frac{2}{3}$$

$$E[X|Y=1] = \text{Prob}(X=1|Y=1) = \frac{0.3}{0.7} = \frac{3}{7}$$

$$\begin{aligned} E[E(X|Y)] &= E[X|Y=0] \times \frac{3}{10} + E[X|Y=1] \times \frac{7}{10} \\ &= \frac{2}{3} \times \frac{3}{10} + \frac{3}{7} \times \frac{7}{10} = \frac{1}{5} + \frac{3}{10} = \frac{5}{10} = \frac{1}{2} \end{aligned}$$

On the other hand,

$$E[X] = \text{Prob}(X=1) = \frac{5}{10} = \frac{1}{2}$$

Q II)

i) Let  $X = [C \ X_2]$ , where  $X_2$  is a  $n \times k-1$  matrix

$$y = C \hat{\beta}_1 + X_2 \hat{\beta}_2 + M_{CX_2} y, \quad (1)$$

$$P_{CX_2} y = [C \ X_2] \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$$

Also, define

$$Z = C \hat{Z}_1 + X_2 \hat{Z}_2 + M_{CX_2} Z \quad \text{where} \quad (2)$$

$$P_{CX_2} Z = [C \ X_2] \begin{bmatrix} \hat{Z}_1 \\ \hat{Z}_2 \end{bmatrix}$$

From eq (2),

$$M_C Z = M_C X_2 \hat{Z}_2 + M_{CX_2} Z$$

$$M_C (y - C) = M_C X_2 \hat{Z}_2 + M_{CX_2} (y - C)$$

$$M_L (L \hat{\beta}_1 + X_2 \hat{\beta}_2 + M_{LX_2} y) = M_L X_2 \hat{\beta}_2 + M_{LX_2} y$$

$$M_L X_2 \hat{\beta}_2 = M_L X_2 \hat{\beta}_2$$

Let  $K = M_L X_2 = \begin{bmatrix} x_{11} - \frac{\sum_{i=1}^n x_{i1}}{n} & x_{21} - \frac{\sum_{i=1}^n x_{i2}}{n} & \dots \\ \vdots & \vdots & \ddots \\ x_{n1} - \frac{\sum_{i=1}^n x_{i1}}{n} & \dots & \dots \end{bmatrix}_{n \times k}$

So, if  $X_2$  is full rank so is  $K$

$$(K^T K)^{-1} K^T K \hat{\beta}_2 = (K^T K)^{-1} K^T K \hat{\beta}_2$$

$$\hat{\beta}_2 = \hat{\beta}_2$$

In terms of  $\hat{\beta}_1$  and  $\hat{\beta}_2$ , substitute (1) into (2) and note that  $\hat{\beta}_2 = \hat{\beta}_2$  and  $M_{LX_2} Z = M_{LX_2} y$

Then

$$L \hat{\beta}_1 + X_2 \hat{\beta}_2 - c = L \hat{\beta}_1 + X_2 \hat{\beta}_2$$

$$L (\hat{\beta}_1 - c) = L \hat{\beta}_1$$

$$(L^T L)^{-1} L^T L (\hat{\beta}_1 - c) = (L^T L)^{-1} L^T L \hat{\beta}_1$$

$$\hat{\beta}_1 = \hat{\beta}_1 - c$$

ii) Note that  $A = c I_{n \times n}$  for  $c \in \mathbb{R}$ , then

$$P_X y = X \hat{\beta}$$

$$c I P_X y = c I X \hat{\beta}$$

$$P_X c y = c X \hat{\beta}$$

$$P_X z = X c \hat{\beta}$$

$$X \hat{z} = X c \hat{\beta}$$

$$(X^T X)^{-1} X^T X \hat{z} = (X^T X)^{-1} X^T X c \hat{\beta}$$

$$\hat{z} = c \hat{\beta}$$

QIII)

$$i) \quad A = \begin{bmatrix} 1 & 0 & 2 \\ -2 & 1 & -3 \\ 0 & 4 & 5 \end{bmatrix}$$

It is a square matrix of full rank,  
so it is invertible

$$\det A = 1(5+12) + 2(-8) = 17 - 16 = 1$$

$$A^{-1} = \begin{bmatrix} 17 & 8 & -2 \\ 10 & 5 & -1 \\ -8 & -4 & 1 \end{bmatrix}$$

ii)

We know that

$$\begin{aligned} P_Z &= Z(Z^T Z)^{-1} Z^T \\ &= XA(A^T X^T X A)^{-1} A^T X^T \\ &= XA A^{-1} (X^T X)^{-1} (A^T)^{-1} A^T X^T \\ &= X(X^T X)^{-1} X^T = P_X \end{aligned}$$

Therefore,

$$X\hat{\beta} = P_X y = P_Z y = Z\hat{z}$$

In terms of residuals

$$M_X y = (I - P_X) y = (I - P_Z) y = M_Z y$$

iii)

We know :

$$X\hat{\beta} = XA\hat{z}$$

$$(X^T X)^{-1} X^T X \hat{\beta} = (X^T X)^{-1} X^T X A \hat{z}$$

$$\hat{\beta} = A \hat{z}$$

$$\begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ -2 & 1 & -3 \\ 0 & 4 & 5 \end{bmatrix} \begin{bmatrix} \hat{z}_1 \\ \hat{z}_2 \\ \hat{z}_3 \end{bmatrix}$$

$$\hat{\beta}_1 = \hat{z}_1 + 2\hat{z}_3$$

Also,

$$\begin{bmatrix} 17 & 8 & -2 \\ 10 & 5 & -1 \\ -8 & -4 & 1 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = \begin{bmatrix} \hat{z}_1 \\ \hat{z}_2 \\ \hat{z}_3 \end{bmatrix}$$

$$\hat{z}_1 = 17 \hat{\beta}_1 + 8 \hat{\beta}_2 - 2 \hat{\beta}_3$$

## Question IV: Empirical Application

- (a) The following table replicates Table 2.2.1 in *Mostly Harmless Econometrics* using the Labour Force Survey

Table 1: Comparison of individual characteristics BA vs. no-BA

| Variable        | No BA                 | BA                   | Diff                 |
|-----------------|-----------------------|----------------------|----------------------|
| Ln(wage)        | 3.428***<br>(0.001)   | 3.693***<br>(0.001)  | 0.265***<br>(0.001)  |
| tenure (months) | 100.670***<br>(0.140) | 91.100***<br>(0.174) | -9.570***<br>(0.225) |
| female          | 0.461***<br>(0.001)   | 0.571***<br>(0.001)  | 0.110***<br>(0.001)  |
| immigrant       | 0.181***<br>(0.001)   | 0.334***<br>(0.001)  | 0.152***<br>(0.001)  |
| Age 30-34       | 0.126***<br>(0.001)   | 0.163***<br>(0.001)  | 0.037***<br>(0.001)  |
| Age 35-39       | 0.130***<br>(0.001)   | 0.167***<br>(0.001)  | 0.037***<br>(0.001)  |
| Age 40-44       | 0.133***<br>(0.001)   | 0.153***<br>(0.001)  | 0.020***<br>(0.001)  |
| Age 45-49       | 0.130***<br>(0.001)   | 0.130***<br>(0.001)  | 0.001<br>(0.001)     |
| Age 50-54       | 0.129***<br>(0.001)   | 0.111***<br>(0.001)  | -0.018***<br>(0.001) |
| Age 55-59       | 0.123***<br>(0.001)   | 0.083***<br>(0.001)  | -0.041***<br>(0.001) |
| Age 60-64       | 0.112***<br>(0.001)   | 0.054***<br>(0.000)  | -0.058***<br>(0.001) |

**Note:** This table displays average characteristics for individuals with and without a BA, defined as a four year University degree. The source are author's own calculations from the Canadian Labour Force Survey

On average, workers with at least four-year BA earn a statistically significant higher hourly wage. In particular, workers with four-year BA or higher earn 0.265 log points, or 30.34% higher hourly wage. From the table it is clear that average workers across education levels are also different and that these differences are correlated with hourly wages. For instance, the proportion of females and immigrants are higher among workers with a BA, characteristics that are negatively correlated with earnings. On the other hand, tenure is, on average, lower for workers with a BA or higher, which correlates positively with earnings.

- (b) The following table replicates Table 2.2.2 in *Mostly Harmless Econometrics* using the Labour Force Survey

Table 2: Returns to Finishing a four-year university degree

|               | (1)                 | (2)                 | (3)                 | (4)                  | (5)                  |
|---------------|---------------------|---------------------|---------------------|----------------------|----------------------|
| 4-year Degree | 0.265***<br>(0.001) | 0.270***<br>(0.001) | 0.272***<br>(0.001) | 0.289***<br>(0.001)  | 0.305***<br>(0.001)  |
| tenure        |                     |                     | 0.001***<br>(0.000) | 0.001***<br>(0.000)  | 0.001***<br>(0.000)  |
| female        |                     |                     |                     | -0.151***<br>(0.001) | -0.152***<br>(0.001) |
| immigrant     |                     |                     |                     |                      | -0.103***<br>(0.001) |
| Age Group FE  | No                  | Yes                 | Yes                 | Yes                  | Yes                  |
| Obs.          | 5.64e+05            | 5.64e+05            | 5.64e+05            | 5.64e+05             | 5.64e+05             |

**Note:** This table displays the returns to finishing a four-year degree or higher. The source are author's own calculations from the Canadian Labour Force Survey

Adding controls increase the estimated effect of education on workers' earnings. From the previous table, it is clear that the estimated wage increases when controlling for female and immigrant as they are positively correlated with education; and they are negatively correlated with earnings. The effect of adding tenure in the regression is surprising as one will expect that, given the results of part (a), it will reduce the estimated coefficient. However, the table in part (a) does not includes age fixed effects when calculating differences in tenure across the two groups of workers (no BA vs. BA). Therefore, it could be the case that after controlling for age the tenure of workers with BA is larger.

The coefficients estimates are not causal. There are many unobserved differences between workers with no BA and BA that are correlated with earnings, such as unobserved ability. Therefore, the previous regression suffers from omitted variable bias.

- (c) Selection Bias with homogeneous effects of education is:

$$\text{Selection Bias} = \rho \cdot \{\mathbb{E}[a_i | x_i, Univ_i = 1] - \mathbb{E}[a_i | x_i, Univ_i = 0]\}$$

QV)

$$M_{X_1} M_X = (I - P_{X_1}) (I - P_X)$$

$$= I - I P_X - P_{X_1} I + P_{X_1} P_X$$

$$= I - P_X - P_{X_1} + (P_X P_{X_1})^T$$

$$= I - P_X - P_{X_1} + P_{X_1} = (I - P_X) = M_X$$

Also

$$M_X M_{X_1} = (M_X^T M_{X_1}^T)^T$$

$$= (M_{X_1} M_X)^T$$

$$= M_X^T = M_X$$

Projections take a vector and project it into a subspace formed by vectors in the projection. When the column vectors of a matrix belong to that subspace, the projection does not change the matrix.